

PRARABDHA SONI

Senior Software Engineer (SDE-3)

prarabdha21@gmail.com | +91-8118898113 | www.linkedin.com/in/prarabdha-soni

Education

B.TECH in CSE (Computer Science and Engineering) - IIT Guwahati

Sep'21

Senior Software Engineer

With over 3 years of experience in advanced backend technologies including Spring Boot, Node.js, Apache Kafka, Docker, Redis, FastAPI, python, GenAI, TensorFlow and AWS services like Cognito etc

- **Leadership Role:** Spearheading an operational systems pod, guiding a backend team of 5 members, from October 2023 to the present.
- **Project Management:** Led integration projects for third-party systems such as BBPS, E-NACH, and E-SIGN, from May 2023 to October 2024.

Work Experience

1) **Project Name: SaaS Product**

Company Name: Arthmate(Omnifi)

Core Infrastructure

Oct'23 – PRESENT

Backend, team size – 8

Java, Spring Boot, Kafka, Redis, AWS, (Gateway, Cognito, etc.), PostgreSQL, MySQL MongoDB, Swagger

- **Automated Task Manager:** Architected an advanced task manager leveraging recursive **Apache Kafka** event processing. Fully configurable via a Master Data Management (**MDM**) service, automating mission-critical workflows such as **loan disbursement**.
- **Factory Design Pattern Implementation:** Designed a factory-based runtime instantiation mechanism to dynamically retrieve and instantiate beans based on **real-time conditions**, ensuring adaptability and scalability.

Credit Risk Modeling

Feb'24 – PRESENT

Risk team size – 7

Python, XGBoost, Scikit-Learn, FastAPI, AWS Lambda, Graph Neural Networks (GNNs), TensorFlow, LlamaIndex

- Developed a real-time Credit Risk Model using **XGBoost & LlamaIndex**, integrating CIBIL, CRIF, and Experian credit scores for loan default prediction and fraud detection.
- Engineered feature pipelines for borrower risk assessment, including credit history, debt-to-income ratio (**DTI**), loan stacking behavior, and transactional anomalies.
- Built a Credit Decisioning API with **FastAPI** and AWS **Lambda**, reducing loan approval time by 60% while ensuring compliance with RBI guidelines.
- Designed an AI-driven Fraud Detection System leveraging Graph Neural Networks (**PyTorch Geometric, DGL**) to analyze borrower relationships, flagging synthetic identity fraud and collusion networks.

2) **Project Name: LMS/LOS Origin Product**

Company Name: Arthmate

Core Infrastructure

Feb'22 – Oct'23

Backend, team size – 15

Java, Spring Boot, Spring batch, Node.js, Express.js, MongoDB, MySQL, Jest, Mocha, Docker, BullMQ, Redis, Passport.js

- **MSME Financial Systems Optimization:** Enhanced financial systems with the USP of a validator microservice, ensuring that every loan application is flagged as a deviation until all background checks, including Udyam verification, are completed using **Bull messaging queue**.
- **BBPS :** Integrated Bharat Bill Payment System (BBPS) using **serverless** architecture with Cognito ID for security. The system was separated for various banks used by apps like **PhonePe**, with AWS Lambda used to calculate real-time pending EMI figures during transactions.
- **Banking Integrations:** Developed secure integrations with major banks (Axis, ICICI, RBL) and connected middleware to credit bureaus (**CIBIL**, CRIF), enhancing underwriting and disbursement processes.
- **Co-Lending Platform Development:** Engineered a sophisticated co-lending platform, with the USP being a **selector API** integrated with the data science team to dynamically choose co-lenders based on the loan applicant's A, B, and C scores.
- **E-NACH System Development:** Designed and maintained E-NACH systems with net banking-based mandates and multiple redirection steps. Ensured that transactions were fully completed and **rolled back** in case of failures.
- **E-Sign Authorization System:** Developed secure E-Sign systems, using Aadhaar-based **UIDAI** authentication for electronic authorizations, improving transaction efficiency and security.
- **Credit Scoring System:-** Designed and implemented a dynamic credit scoring system that selects the appropriate external scoring provider (such as **CIBIL**, CRIF, or Experian) based on the partner and real-time requirements.